such as its Elastic Compute Cloud. GCP (Google Cloud Platform) offers storage and computer services through Google Compute Engine (GCE), as does Microsoft Azure.

There are many other smaller or niche players in the IaaS market, including Rackspace Managed Cloud, Century Link Cloud and Digital Ocean.

Benefits of IaaS

IaaS has a number of benefits, some of which are:

- Scalability. Resources are available as and when required by the customer and, hence, there are no delays in expanding capacity or wastage of any used capacity.
- Saves time and cost.
- Utility-based billing. The service can be accessed on demand and the user pays only for the resources that he or she uses.
- Location independence. The service can be accessed from anywhere using the Internet, so long as the security protocol of cloud allows it.
- More physical security.
- No single point of failure.
- Rapid innovation. Suppose an entrepreneur wants to launch a new product, the necessary computing infrastructure is readily available within hours in IaaS—it can sometimes takes months using other cloud platforms.
- Increases stability, reliability and supportability.
- Focuses on core business rather than IT infrastructure.

IaaS provides high-performance computing on supercomputers, computer grids or computer clusters. This helps solve complex problems involving millions of variables or calculations. Examples include earthquake and protein folding simulations, climate and weather predictions, financial modelling and product design evaluation.

Further, IaaS economically provides analysis of Big Data. Mining data sets (database includes cloud-based SQL and NoSQL) to locate or flush out hidden patterns requires huge amounts of processing power, which can be potentially done on an IaaS platform.

Running websites using IaaS can be less expensive than traditional Web hosting. IaaS provides features such as content delivery networks and load balancing. HP Enterprise Converged infrastructure also provides IaaS, and is a part of converged cloud solutions for public, private or hybrid clouds.

Some IaaS cloud service vendors

- Amazon AWS
- Century Link Cloud
- Cisco Metapod
- Digital Ocean
- Google Compute Engine (GCE)
- Green Cloud Technologies
- HP Enterprise Converged Infra
- IBM Smart Cloud Enterprise
- Linode
- Microsoft Azure
- Rackspace Managed Cloud
- Skytap Agile Development

Technical challenges

According to a recent *Forbes* report, 2.5 quintillion bytes of data are being generated every day, and this is expected to multiply several times in the coming years. All the new technologies like machine learning and the Internet of Things (IoT) are fuelling data creation and, at the same time, increasing vulnerability to data breaches.

Well, the spectre of potential breaches is haunting both small and large businesses. Losing data can affect businesses very hard. For most medium- to large-scale businesses, experiencing a single hour of downtime can cost more than one million dollars. With such potential losses, it is no wonder that enterprises invest heavily in IaaS-based cloud systems.

While IaaS-based cloud computing brings many advantages, this model also raises significant technical security considerations and questions. Among these issues are operational trust modes, resource sharing of attack strategies and digital forensics.

Legal issues

In addition to the identified technical challenges, legal issues associated with IaaS-based cloud computing need to be analysed into a risk analysis plan. That way, potential consumers can make informed decisions about the appropriateness of utilising potential valuable resources. Some legal issues include consideration of jurisdictional problems, understanding and evaluation of cloud, stake-holders' rights, and technical approaches to addressing the associated legal and jurisdictional issues.

IaaS is becoming extremely popular in mobile development, and this is especially the case with startups. It is being used to develop Web and mobile analytical tools, which are designed to oversee source code management, testing, tracking, payment gateways and so on.

System admins face numerous challenges today when it comes to managing IT infrastructure and data. Virtualisation software help them address these issues by making available virtual computer hardware platforms, virtual storage devices and virtual computer network resources. Virtual machines in an IaaS cloud system are helping make use of today's high-powered hardware to the fullest. END

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